DIC09
NAND and NOR Logic Gates

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Follow Lab 9 (PDF pages 35–38): size a reference inverter, then compare two-input NAND/NOR input order, dimensions, transfer curves, noise margins, speed and power.

2 Specification mapping

Handout requirement (PDF page)	Implementation	Verification and scope
Step 1: balance inverter (35,38)	01_inverter_sizing	Ratios 3–7; select 6
Steps 2/3: NAND/NOR sizing and DC (36–38)	02_nand, 03_nor, 04_sizing_refinement	A, B, A=B; complete NM
Questions 1/2: late input and parasitics (38)	One input switches after the other settles	Both input positions measured
Question 3: factors 1.5/2/2.5/3 (38)	Four cases in 02/03	Compare output rise/fall
Question 4: speed, power and area (38)	Average supply power and total widths	Equal loads; area is only a proxy

3 Circuit and design derivation

Length is fixed at $0.25\,\mu$m and reference $W_n = 0.6\,\mu$m. Sweep $W_p/W_n = 3, 4, 5, 6, 7$ and choose 6 as the gate reference. NAND has series NMOS and parallel PMOS; NOR reverses these networks. A is nearest the output in the NAND stack and nearest the supply in the NOR stack. Each gate retains simultaneous A=B switching and the two individually sensitized inputs. The inactive input is held at 1 for NAND and 0 for NOR.

Initially, series widths are multiplied by K=1.5/2/2.5/3 to test $R \propto L/W$. Input-specific refinement then compensates for parasitic effects. The selected third refinement uses NAND PMOS A/B widths 4.14/5.04 and NMOS widths 1.5 each; NOR PMOS widths are 21.6 each and NMOS A/B widths

2.7/1.02, in μm. The source Lab9 first file incorrectly referenced an external PWM circuit; a self-contained inverter calibration replaces it. All gates include their own circuitry and measurements.

4 Simulation setup

This teaching rewrite uses a separately supplied model at 1.0 V and 25 °C. The specified topology, loads and comparative dimensions are retained. The handout's 2.5 V process results are not acceptance values for this model. The chosen 1 V is a simulation condition, not a verified device rating. MOS bulk terminals are explicitly connected to the corresponding supply rails.

HSPICE S-2021.09 uses RUNLVL=6, retaining waveforms, printed data and native measurements. Figures are redrawn from actual data; CSV files are in each Lab's results directory. Successful solution does not establish process reliability. Warnings from encrypted model blocks remain suppressed; the existing diagnostic did not expose actionable further details, so the warnings are not declared harmless and complete model-limit compliance is not claimed.

Every output has 20 fF. Input edges are 0.1 ns, period 5 ns; simulate 15 ns with 1 ps printing. DC is 0–1 V in 0.5 mV steps. Supply power is averaged over full cycles from 5–15 ns and excludes ideal input-source energy. NM uses the Lab8 boundary-output definition. The handout specifies approximate equality without a tolerance; 10% is an added engineering comparison target, not a claimed original acceptance threshold.

5 Results and specification checks

All four decks and 16 cases completed without failed measurements.

W_p/W_n	V_M (V)	t_r (ps)	t_f (ps)
3	0.4825	222.64	110.85
4	0.4956	173.83	113.49
5	0.5058	144.13	115.26
6	0.5142	124.85	117.32
7	0.5212	110.93	119.71

Gate	Input	V_M (V)	NM_L (V)	NM_H (V)
NAND	both	0.5464	0.4615	0.3621
NAND	a	0.4923	0.3924	0.4226
NAND	b	0.4827	0.3808	0.4331
NOR	both	0.4792	0.3849	0.4430
NOR	a	0.5455	0.4539	0.3585
NOR	b	0.5352	0.4442	0.3712

Selected design	t_r (ps)	t_f (ps)	t_{pHL} (ps)	t_{pLH} (ps)
INV	123.30	116.03	109.74	96.30
NAND A	130.26	111.14	99.02	102.50
NAND B	124.24	111.18	107.10	104.32
NOR A	134.89	122.00	135.99	121.49
NOR B	128.66	116.35	150.46	79.97

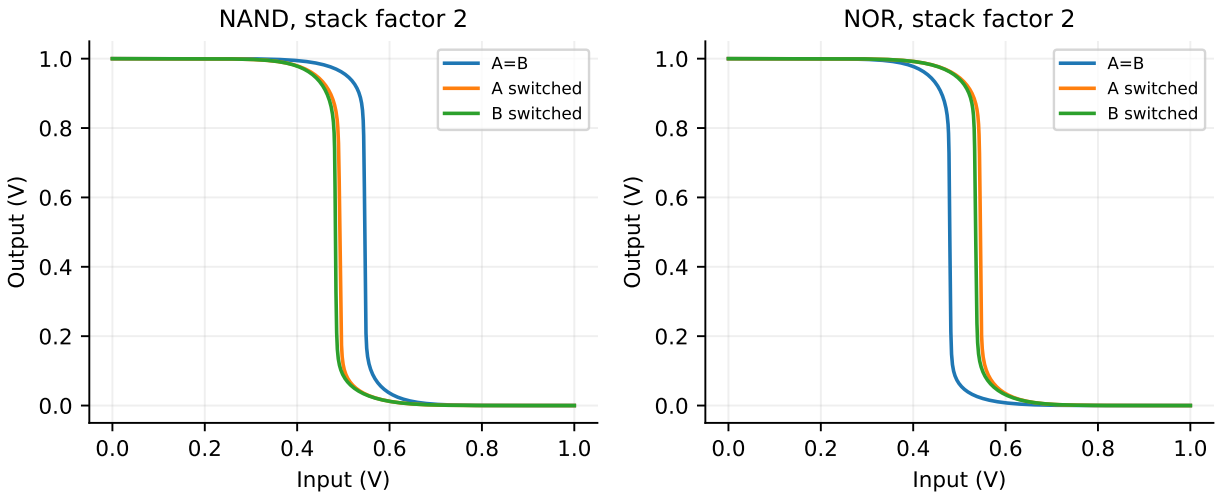


Figure 1: Three input conditions with stack factor 2; redrawn from this run's HSPICE data.

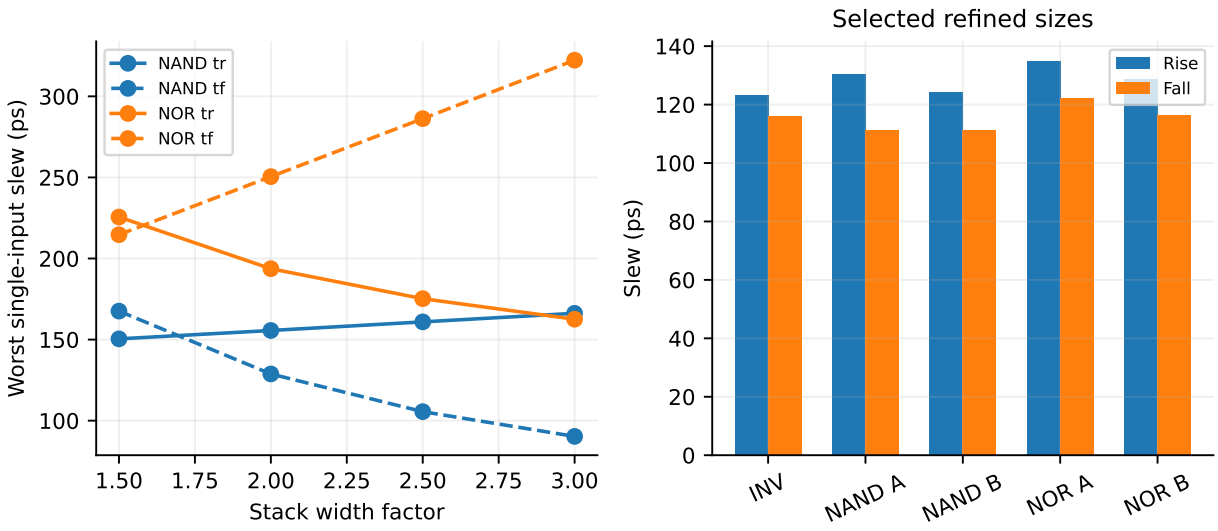


Figure 2: Stack-factor sweep and final size refinement; redrawn from this run's HSPICE data.

Mean supply power and total widths at initial K=2:

Gate	$P_{A=B}$ (μW)	P_A (μW)	P_B (μW)	$\sum W$ (μm)
NAND	7.145	6.480	7.299	9.6
NOR	9.134	12.916	7.232	15.6

6 Discussion

The ratio-6 inverter has a threshold near 0.514 V and rise/fall times of about 124.85/117.32 ps: close, not perfectly symmetric. In the selected refinement, the maximum single-input slew deviation from the same-deck inverter is about 9.4%. Simultaneous activation of parallel devices is faster, so equality across every input mode is not expected.

The latest signal should drive the series device nearest the output: A for NAND, B for NOR. The other input is held settled in this test. Internal-node charging and body effect make simple stack-count width multiplication inexact; widening also increases junction, overlap and input capacitance. At initial K=2, total widths are 9.6 for NAND and 15.6 μm for NOR. This is an active-width proxy at equal L, excluding layout, contacts and routing. CSV files retain average supply power for every input mode. NOR is generally slower and more energy-consuming in this equal-load comparison, but no universal cross-process ranking is claimed. Matching slew by greatly enlarging NOR trades area and input loading for speed.

7 Conclusion

All three steps and four questions have corresponding tests. The specified multipliers are retained before further refinement. Approximate 10% matching is a result against this run's added engineering target; original-process values and unknown reliability constraints are not declared satisfied.